
FusedSpaceFed: Enhanced Federated Learning Through Dual-Space Data Fusion

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Abstract

We introduce FusedSpaceFed, a federated learning framework designed to mitigate client drift under heterogeneous data distributions through dual-space data fusion. In settings such as medical imaging, client heterogeneity may reflect both different label distributions and site-specific acquisition characteristics. FusedSpaceFed maintains a private convolutional encoder for each client, while the decoder and classifier are shared through federated aggregation. At each client, the original input is combined with the reconstruction produced by the local encoder and the shared decoder, and the resulting fused representation is used for classification. Training alternates between an encoder-only reconstruction warm-up and an end-to-end classification phase. We provide an exact fixed-round decomposition of classifier-parameter gradient dispersion, which separates the dispersion observed without fusion from the variance and covariance terms introduced by the fused representation. The decomposition identifies the variance-covariance condition under which fusion lowers cross-client gradient dispersion at a fixed model state. Experiments show improved predictive performance under severe non-IID partitions and lower empirical classifier-gradient dispersion than the considered federated baselines.

1 INTRODUCTION

Federated Learning (FL) enables multiple clients to train a shared model without sharing their private data [27]. It is particularly suited to medical imaging, finance and mobile devices, where privacy regulations or communication constraints limit centralized data collection [17]. In practice, heterogeneous client data distributions can hinder performance by inducing divergent local updates and slowing convergence.

A challenge arises under non-IID data, where client gradients may deviate from their mean. Local and distributed SGD analyses show how statistical heterogeneity biases local updates and slows optimization [33, 35], potentially requiring conservative step sizes, with convergence rates degrading with the degree of heterogeneity. Existing approaches address this through proximal regularization, control variates, gradient tracking or personalized components [3, 5, 13, 18, 22–24].

FusedSpaceFed instead modifies the classifier input by combining the original image with a reconstruction from a private encoder and federated decoder. The objective is to reduce the dispersion of client updates induced by heterogeneous local data.

As a guiding example, consider medical imaging federation, where heterogeneity extends beyond label imbalance [12, 32]. Clinical sites may serve populations that differ in age, comorbidities and environmental exposures. These differences can affect both disease prevalence and image characteristics in the same diagnostic category. Moreover, clinical sites rely on different scanners, acquisition protocols, reconstruction pipelines and calibration procedures [10, 11]. Thus, two images of the same pathology acquired at different hospitals may differ in intensity, contrast and noise characteristics. These instrumental variations introduce systematic site-specific biases unrelated to the underlying medical content, yet strongly affect model training.

Consequently, gradients computed at different sites

may diverge because each client observes the data through a different “instrument”. From an optimization perspective, this effect can amplify client drift and impair convergence and generalization.

FusedSpaceFed addresses heterogeneity through the representation supplied to the classifier. Each client retains a private multiscale encoder E_i , while the decoder D and classifier are synchronized by the server. The classifier processes $x + D(E_i(x))$, combining the original image with a reconstruction conditioned on the client. The shared decoder is influenced by all participating clients, while the private encoder preserves client-specific adaptation.

Our contributions are:

- an asymmetric architecture with private multiscale encoders, a shared decoder and a shared classifier;
- local training that alternates encoder-only reconstruction warm-up with classification-only end-to-end updates of all three components;
- an exact fixed-round decomposition of classifier-gradient dispersion into original dispersion, fusion-induced variance and cross-covariance;
- evaluation on twelve MedMNIST datasets, pathological partitions, Dirichlet stress tests, natural writer-level FEMNIST and five-domain Digits.

The architectural contribution lies in the coordinated combination of a private multiscale encoder, a federated decoder, additive input-space fusion and per-round two-phase optimization. Together, these components construct a client-conditioned representation while transferring information through the shared reconstruction and classification pathways.

2 RELATED WORK

Federated optimization must address statistical heterogeneity and partial participation. FedAvg alternates local optimization and server-side parameter averaging [27]. FedProx penalizes deviations from global parameters [23], while SCAFFOLD uses client and server control variates to correct local drift [18]. FedDyn adds dynamic regularization, MOON aligns local and global representations contrastively, and FedNTD distills predictions for non-ground-truth classes [1, 20, 21].

Personalized FL separates shared and private components. FedPer keeps personalization layers local and federates a common base [3]; FedRep shares a representation with private prediction heads [5];

Ditto trains global and regularized client-specific models [22]. FedBN retains local batch-normalization components to address feature shifts, including medical acquisition shifts [24]. FusedSpaceFed uses a different split: private multiscale encoders, a shared decoder and classifier, and additive fusion of the reconstructed and original inputs.

Whereas MOON and FedNTD add representation- or prediction-level objectives, FusedSpaceFed constructs client-conditioned inputs through architecture and reconstruction warm-up, without a representation-alignment penalty. Related structural interventions have been studied in centralized learning. GloNets exposes the network head to features from different depths [6]. Dendrite-activated and multi-bias units modify preactivation patterns to preserve information flow and improve parameter efficiency [28–30]. SwitchPath stochastically exposes alternative activation pathways during training [8, 9]. These works illustrate how computational pathways can shape representation learning alongside the training objective.

Generative FL exchanges information without sharing raw samples. FedCVAE synthesizes representative data with conditional variational autoencoders [15]; SDA-FL generates, pseudo-labels and redistributes synthetic datasets [25]; FissionVAE splits generative components across client clusters [16]. FusedSpaceFed instead keeps its reconstruction pathway active during both training and inference, adding its output to the original input.

2.1 Federated Gradient Dispersion

Let θ denote the parameters of a shared classifier and let $\mathcal{S}_t$ be the set of clients participating at communication round t , with $m_t = |\mathcal{S}_t|$. For a scalar task loss ℓ_{task} , the local objective of client i is

$$F_i(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\ell_{\text{task}}(\mathcal{C}_\theta(x), y)].$$

All client gradients must be evaluated at a common parameter vector in order for their dispersion to be meaningful. At a reference point θ_t , define

$$g_{i,t} := \nabla_\theta F_i(\theta_t), \quad \bar{g}_t := \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} g_{i,t}.$$

The corresponding gradient dissimilarity is

$$\Gamma_t = \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} \|g_{i,t} - \bar{g}_t\|_2^2. \quad (1)$$

The quantity Γ_t measures the absolute dispersion of client gradients evaluated at the common classifier state θ_t . Lower values indicate smaller squared deviations from the mean client gradient.

3 METHODOLOGY

We describe FusedSpaceFed for medical images, although the architecture is not setting-specific. A convolutional autoencoder supplies fused inputs $x + \tilde{x}$ to a classifier; the encoder remains private, while decoder and classifier are shared. Algorithm 1 summarizes training.

3.1 Dual-Space Architecture

Client $i \in \{1, \dots, K\}$ holds $\mathcal{D}_i = \{(x_j^{(i)}, y_j^{(i)})\}_{j=1}^{n_i}$, with inputs in $\mathcal{X} = \mathbb{R}^{c_{\text{in}} \times H \times W}$.

All tensor dimensions are positive integers; image height H and width W are multiples of four for the encoder’s two pooling stages. The input channel count c_{in} and number of outputs c_{out} depend on the dataset. Labels lie in $\{0, \dots, c_{\text{out}} - 1\}$ for multiclass tasks and $\{0, 1\}^{c_{\text{out}}}$ for multilabel ChestMNIST.

Private multiscale encoder. The private compact U-Net encoder E_i [31] has base width 16 and returns the multiscale tuple

$$E_i(x) = h_i(x) = \left(s_i^{(1)}(x), s_i^{(2)}(x), z_i(x) \right),$$

with skip representations $s_i^{(1)}(x) \in \mathbb{R}^{16 \times H \times W}$ and $s_i^{(2)}(x) \in \mathbb{R}^{32 \times (H/2) \times (W/2)}$, and bottleneck $z_i(x) \in \mathbb{R}^{d_z \times (H/4) \times (W/4)}$. Here d_z counts bottleneck channels. Thus E_i maps $\mathcal{X}$ to the product of these three tensor spaces.

Shared decoder. The shared decoder D maps this product space back to $\mathcal{X}$:

$$\tilde{x}_i = D(h_i(x)) \in \mathcal{X}.$$

Its output layer is linear. Each round initializes local decoder copies from the current shared parameters.

Shared classifier and fusion. The shared classifier $\mathcal{C} : \mathcal{X} \rightarrow \mathbb{R}^{c_{\text{out}}}$ produces logits:

$$x_{\text{fused}}^{(i)} = x + \tilde{x}_i, \quad o_i = \mathcal{C}\left(x_{\text{fused}}^{(i)}\right) \in \mathbb{R}^{c_{\text{out}}}.$$

Fusion preserves the input’s shape. The decoder output and fused input are not constrained to the input’s normalization range.

3.2 Training Objectives

Local optimization consists of two consecutive phases in every communication round. The two losses are not combined into a weighted joint objective.

Phase 1: encoder reconstruction warm-up. At the beginning of round t , client i receives the shared decoder D_{ϕ_t} and classifier $\mathcal{C}_{\theta_t}$. Both components are frozen, and only the private encoder parameters ψ_i are updated for E_w epochs using

$$\mathcal{L}_{\text{rec}}^{(i)} = \mathbb{E}_{x \sim \mathcal{D}_i} \left[\|x - D_{\phi_t}(E_{\psi_i}(x))\|_2^2 \right].$$

Phase 2: end-to-end classification. After warm-up, client i unfreezes its local decoder and classifier and updates the encoder, decoder, and classifier for E local epochs using only

$$\mathcal{L}_{\text{cls}}^{(i)} = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\ell_{\text{task}}(\mathcal{C}_{\theta_i}(x + D_{\phi_i}(E_{\psi_i}(x))), y)].$$

Phase 2 has no pixel-level reconstruction term: gradients of $\mathcal{L}_{\text{cls}}^{(i)}$ pass through the fused input to the classifier, decoder and private encoder. Multiclass MedMNIST and FEMNIST use softmax cross-entropy; ChestMNIST uses per-label binary cross-entropy with logits. Phase 1 uses mean squared error throughout.

The two phases serve complementary objectives: reconstruction adapts each private encoder to the current shared decoder, while classification jointly optimizes the encoder, decoder, and classifier for prediction. The decoder output therefore serves as a task-adapted image-space signal for additive fusion. Checkpoint-based diagnostics on Digits illustrate these distinct roles (Appendix D.1).

3.3 Global Aggregation

Let $\mathcal{S}_t$ denote the active clients at round t and $m_t = |\mathcal{S}_t|$. After local training, every active client sends its updated decoder parameters ϕ_i^{t+1} and classifier parameters θ_i^{t+1} to the server. FusedSpaceFed uses uniform client averaging:

$$(\phi_{t+1}, \theta_{t+1}) = \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} (\phi_i^{t+1}, \theta_i^{t+1}). \quad (2)$$

All methods intentionally average clients uniformly, regardless of sample count. Private encoder parameters ψ_i are neither transmitted nor averaged. Returning clients reuse their stored encoder with the latest shared decoder and classifier.

Algorithm 1 FusedSpaceFed training protocol

```

1: Initialize: shared decoder  $\phi_0$ , shared classifier  $\theta_0$ , and
   one persistent private encoder  $\psi_i^0$  for each client
2: for communication round  $t = 0, \dots, T - 1$  do
3:   Select the active client set  $\mathcal{S}_t$ 
4:   Broadcast  $(\phi_t, \theta_t)$  to all  $i \in \mathcal{S}_t$ 
5:   for each  $i \in \mathcal{S}_t$  in parallel do
6:     Initialize local decoder  $\phi_i \leftarrow \phi_t$  and classifier
        $\theta_i \leftarrow \theta_t$ 
7:     Restore the persistent private encoder  $\psi_i$ 
8:     Freeze  $(\phi_i, \theta_i)$ 
9:     Update only  $\psi_i$  for  $E_w$  epochs using  $\mathcal{L}_{\text{rec}}^{(i)}$ 
10:    Unfreeze  $(\phi_i, \theta_i)$ 
11:    Update  $(\psi_i, \phi_i, \theta_i)$  for  $E$  epochs using only  $\mathcal{L}_{\text{cls}}^{(i)}$ 
12:    Store  $\psi_i$  locally
13:    Send  $(\phi_i, \theta_i)$  to the server
14:  end for
15:   $\phi_{t+1} \leftarrow \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} \phi_i$ 
16:   $\theta_{t+1} \leftarrow \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} \theta_i$ 
17: end for
    
```

3.4 Personalized Inference

FusedSpaceFed defines a family of client-conditioned predictors composed of a shared decoder and classifier together with one private encoder for each participating client. For client i , the logit vector is

$$o_i = \mathcal{C}_{\theta_T} \left(x + D_{\phi_T} \left(E_{\psi_i^T}(x) \right) \right).$$

For an unseen client, a new private encoder can be initialized and adapted through reconstruction warm-up with the final shared decoder. We evaluate the client-conditioned predictors of participating clients.

4 DISPERSION ANALYSIS

We analyze how dual-space fusion changes the dispersion of classifier-parameter gradients across clients. At communication round t , the analysis conditions on the current private encoders, shared decoder, and shared classifier, allowing the original-input and fused-input gradient fields to be compared at the same model state.

4.1 Original and Fused Client Objectives

Fix a communication round t , an active client set $\mathcal{S}_t$, the shared classifier parameters θ_t , the shared decoder parameters ϕ_t , and the current private encoder parameters $\psi_{i,t}$. During this analysis, all of these quantities are held fixed except for the argument θ with respect to which classifier gradients are defined. For client i , define the reference original-input objective

$$F_{i,t}^{\text{orig}}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\ell_{\text{task}}(\mathcal{C}_{\theta}(x), y)]$$

and the fused-input objective

$$F_{i,t}^{\text{fused}}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\ell_{\text{task}}(\mathcal{C}_{\theta}(x + D_{\phi_t}(E_{\psi_{i,t}}(x))), y)].$$

Both are scalar loss functions, and their gradients are taken with respect to the same classifier parameter vector:

$$g_{i,t}^s = \nabla_{\theta} F_{i,t}^s(\theta_t), \quad s \in \{\text{orig}, \text{fused}\}.$$

Define the fusion-induced gradient difference $b_{i,t} = g_{i,t}^{\text{fused}} - g_{i,t}^{\text{orig}}$. Let

$$\bar{g}_t^s = \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} g_{i,t}^s, \quad s \in \{\text{orig}, \text{fused}\},$$

$$\bar{b}_t = \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} b_{i,t}.$$

where $m_t = |\mathcal{S}_t|$.

4.2 Exact Fixed-Round Decomposition

The original and fused classifier-gradient dispersions are

$$\Gamma_t^s = \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} \|g_{i,t}^s - \bar{g}_t^s\|_2^2, \quad s \in \{\text{orig}, \text{fused}\}.$$

Proposition 1 (Fixed-round gradient-dispersion identity). *At any fixed round t ,*

$$\Gamma_t^{\text{fused}} = \Gamma_t^{\text{orig}} + B_t + \Phi_t. \quad (3)$$

where

$$B_t = \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} \|b_{i,t} - \bar{b}_t\|_2^2 \geq 0,$$

$$\Phi_t = \frac{2}{m_t} \sum_{i \in \mathcal{S}_t} \langle g_{i,t}^{\text{orig}} - \bar{g}_t^{\text{orig}}, b_{i,t} - \bar{b}_t \rangle.$$

Proof. Because

$$g_{i,t}^{\text{fused}} - \bar{g}_t^{\text{fused}} = (g_{i,t}^{\text{orig}} - \bar{g}_t^{\text{orig}}) + (b_{i,t} - \bar{b}_t),$$

expanding the squared Euclidean norm and averaging over $i \in \mathcal{S}_t$ gives

$$\begin{aligned}
 \Gamma_t^{\text{fused}} &= \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} \|g_{i,t}^{\text{orig}} - \bar{g}_t^{\text{orig}}\|_2^2 \\
 &\quad + \frac{1}{m_t} \sum_{i \in \mathcal{S}_t} \|b_{i,t} - \bar{b}_t\|_2^2 \\
 &\quad + \frac{2}{m_t} \sum_{i \in \mathcal{S}_t} \langle g_{i,t}^{\text{orig}} - \bar{g}_t^{\text{orig}}, b_{i,t} - \bar{b}_t \rangle \\
 &= \Gamma_t^{\text{orig}} + B_t + \Phi_t.
 \end{aligned}$$

□

Corollary 1 (Fixed-round reduction condition). *At the fixed state considered above,*

$$\Gamma_t^{\text{fused}} < \Gamma_t^{\text{orig}} \iff B_t + \Phi_t < 0.$$

The term B_t measures the additional cross-client variability introduced by the fusion-induced gradient differences. The cross-term Φ_t is negative when these differences tend to oppose the original client deviations from their mean. Therefore, fusion is associated with lower dispersion only when the negative cross-covariance is sufficiently large to compensate for B_t .

Can learned fusion satisfy the reduction condition in Equation (3)? On Digits, we compare original-input and fused-input classifier gradients at identical model states. At the final checkpoints, $B_t + \Phi_t < 0$ holds in four of five seeds with frozen BatchNorm statistics and all five with batch statistics, illustrating the reduction condition for absolute dispersion. See Appendix D.3 for the decomposition and normalized and directional diagnostics.

4.3 Communication Complexity

Let P_C and P_D denote the numbers of transmitted scalar values in the classifier and decoder states, respectively, and let $m_t = |\mathcal{S}_t|$. At round t , the server sends one classifier and one decoder state to every active client, and every active client returns the corresponding updated states. The resulting bidirectional communication is $\text{Comm}_t^{\text{FusedSpaceFed}} = 2m_t(P_C + P_D)$ transmitted scalar values, or $2m_t(P_C + P_D)b$ bytes, when each scalar occupies b bytes.

The private encoder is neither uploaded nor downloaded. Relative to FedAvg and FedProx using the same classifier, FusedSpaceFed therefore communicates an additional $2m_tP_D$ scalar values per round. For the convolutional decoder, P_D is determined by the channel widths, convolutional kernels, upsampling layers, and skip-connection processing blocks. SCAFFOLD additionally maintains client and server control variates, with communication determined by the specific control-variate implementation.

5 EXPERIMENTAL EVALUATION

We evaluate FusedSpaceFed on a diverse set of medical imaging tasks [36], a standardized suite of biomedical datasets spanning multiple modalities (X-ray, OCT, CT, fundus, dermatoscopy, microscopy), and FEMNIST [4], a popular benchmark for a federated learning scenario.

For the main MedMNIST experiments, we distribute the official training split across 10 clients using a class-wise Dirichlet allocation with $\alpha \in \{0.05, 0.1, 0.2, 0.5, 1.0\}$. Lower values of α produce more concentrated client label distributions. Every official training example is assigned to exactly one client. We compare FusedSpaceFed with three estab-

lished federated learning reference methods:

- **FedAvg** [27]: Standard baseline using parameter averaging.
- **FedProx** [23]: Adds a proximal term to stabilize local updates.
- **SCAFFOLD** [18]: Uses control variates to correct client drift.

All MedMNIST methods use the same ResNet20-v2 [14] classifier, initialization and client partition within each independent run. FusedSpaceFed also uses the compact U-Net of Section 3.

MedMNIST images are loaded from the official 64×64 release and resized to 32×32 . Training uses 50 communication rounds, 3 local classification epochs per round, and full participation of all medical clients. FusedSpaceFed performs one additional encoder warm-up epoch at the beginning of each round. The MedMNIST batch size is 128. The classifier is optimized using SGD with learning rate 0.01, while the encoder and decoder are optimized using Adam [19] with learning rate 0.001. FedProx uses proximal coefficient $\mu = 0.01$. Client states are aggregated using a uniform client average.

Results are averaged over five independent runs with different seeds. The official test split is evaluated only after the final communication round. The full implementation, data-partitioning procedure, and experiment configurations are available in the supplementary material. For MedMNIST, FedAvg, FedProx and SCAFFOLD are evaluated using their final shared classifier. For FusedSpaceFed, each of the 10 private client encoders is combined with the final shared decoder and classifier and evaluated on the official test split. We report the uniform mean across these 10 client-conditioned pipelines.

The MedMNIST evaluation measures the generalization of each client-conditioned predictor to a common test distribution after federated training on heterogeneous local class distributions. Using the same official test split for all methods provides a common evaluation target across the different training partitions. Each private encoder operates together with a shared classifier trained through updates from the participating clients, while additive fusion retains the original input as a direct classification pathway. The uniform average summarizes performance across the ten client-conditioned predictors. For FEMNIST, evaluation instead follows the natural client structure, using each writer’s encoder on that writer’s test examples.

We select $d_z \in \{8, 16, 32, 64\}$ separately for each MedMNIST dataset using FusedSpaceFed accuracy on

the official validation split. The selected width is fixed across all test evaluations and heterogeneity settings for that dataset (Table 1).

ChestMNIST is multi-label. To construct its Dirichlet client partition, each training example is assigned a single proxy stratum: the first positive label is used when at least one finding is present, and samples without any positive finding are assigned to an additional “no finding” stratum. This proxy is used only for partitioning. The original multi-label target vector is retained for training and evaluation, which use binary cross-entropy with logits. For ChestMNIST, predictions are obtained by thresholding each sigmoid output at 0.5, and accuracy denotes the mean element-wise agreement across samples and labels.

5.1 Performance Across Medical Datasets

Table 1 summarizes the average test accuracy across datasets under two different Dirichlet settings. Under strong heterogeneity ($\alpha = 0.05$), FusedSpaceFed outperforms all baselines on every dataset, with gains of 0.12–7.32 percentage points over the best baseline. Under moderate heterogeneity ($\alpha = 0.50$), FusedSpaceFed achieves the highest average accuracy on nine of twelve datasets.

Across the intermediate values of α , accuracy generally increases as the expected client label distributions become less concentrated, with minor non-monotonic fluctuations across configurations. At $\alpha = 1$, training remains federated, but the expected class allocations are more balanced and the performance differences among the evaluated methods become smaller.

We additionally compute the empirical classifier-parameter gradient dissimilarity defined in Equation (1). For each method, the measurement is performed at the final trained model state using one deterministic training mini-batch per medical client. Gradients are taken only with respect to the ResNet20-v2 classifier parameters. For FusedSpaceFed, each batch is first transformed using the corresponding private encoder and the final shared decoder. Table 2 reports the mean squared distance between each client gradient and the mean client gradient. The resulting value measures absolute cross-client classifier-gradient dispersion at each method’s final trained state.

We also evaluate PathMNIST, DermaMNIST, RetinaMNIST and BloodMNIST with each client restricted to two assigned classes and otherwise identical experimental configurations. FusedSpaceFed achieves the highest reported accuracy on all four datasets, with particularly large margins on PathMNIST, DermaMNIST and RetinaMNIST (Table 3), consistent with improved robustness under extreme label skew.

5.2 Stress Test

In addition to the five Dirichlet settings used in the main experiments, we perform a separate stress-test sweep on PathMNIST using $\alpha_j = 0.05 + 0.10j$, $j = 0, \dots, 11$. This twelve-point grid corresponds to the values displayed in Figure 1. This grid spans different levels of label heterogeneity, with lower α values yielding more concentrated client label distributions.

We train the four methods on PathMNIST for every value in the stress-test grid and record test accuracy and classifier-gradient dissimilarity after the final communication round. As shown in Figure 1(a), accuracy decreases for all methods as the client partitions become more concentrated. FusedSpaceFed maintains the highest accuracy in the strongest-skew portion of the sweep, while the differences among methods narrow as α increases. Figure 1(b) shows the corresponding classifier-gradient dispersion. FusedSpaceFed produces substantially lower dispersion in the most heterogeneous settings and remains competitive over the rest of the evaluated range.

5.3 FEMNIST Benchmark

We further evaluate all methods on **FEMNIST** from the LEAF benchmark [4], a widely adopted reference dataset for federated learning. FEMNIST consists of handwritten characters from thousands of distinct writers and exhibits natural client-level heterogeneity, as each client corresponds to a single user with highly skewed label and style distributions. This setting closely reflects realistic federated scenarios and is commonly used to assess robustness to client drift and non-IID data.

We retain LEAF’s natural writer-level partition: each writer is one client, with no pooling or repartitioning. Each round samples a subset of writers for local training. Private FusedSpaceFed encoders and SCAFFOLD client control variates persist when writers re-join.

FedAvg, FedProx and SCAFFOLD use a shared classifier, whereas FusedSpaceFed combines the shared decoder and classifier with writer-specific private encoders. We therefore report writer-conditioned personalized test performance: the test examples of each writer are processed using that writer’s private encoder together with the final shared decoder and classifier.

Gradient dissimilarity is evaluated after the final round using one deterministic batch from each of 50 seeded writers and gradients with respect to the classifier parameters only. Table 4 reports the mean results over five independent runs. FusedSpaceFed obtains the highest writer-conditioned accuracy and the low-

Table 1: Average accuracy across datasets under strong ($\alpha = 0.05$) and moderate ($\alpha = 0.50$) heterogeneity. Fused denotes FusedSpaceFed; the validation-selected d_z is common to both settings.

Dataset	$\alpha = 0.05$				$\alpha = 0.50$				d_z
	FedAvg	FedProx	SCAFFOLD	Fused	FedAvg	FedProx	SCAFFOLD	Fused	
Path	53.45	48.11	39.90	60.77	90.49	90.70	88.38	91.68	16
Chest	90.28	89.93	89.44	93.22	94.25	93.03	93.65	95.25	32
Derma	67.70	65.94	63.35	69.60	72.19	72.20	68.64	72.55	16
OCT	41.56	42.50	38.69	46.25	67.72	70.23	67.69	69.88	8
Pneumonia	69.12	70.24	70.01	72.95	82.26	81.60	82.79	83.42	16
Retina	48.13	49.69	48.75	53.00	60.13	61.23	61.97	62.60	64
Breast	80.25	79.33	77.78	81.61	85.35	83.34	83.43	85.70	32
Blood	53.87	47.38	45.68	58.83	90.43	86.51	87.00	90.18	16
Tissue	35.65	33.28	34.74	38.21	61.65	59.56	60.89	64.33	32
OrganA	23.65	25.12	19.72	26.01	81.27	81.88	80.44	82.23	16
OrganC	22.81	24.64	18.96	24.76	79.42	77.51	79.50	80.16	32
OrganS	22.55	21.96	20.12	24.11	69.56	69.17	67.50	69.45	32

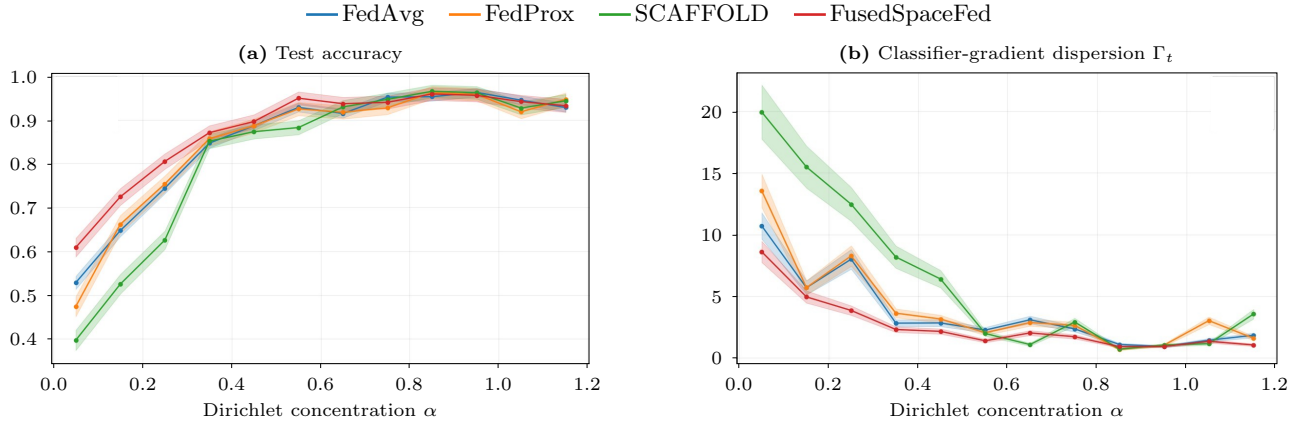


Figure 1: PathMNIST stress test: (a) test accuracy and (b) final-round classifier-gradient dispersion Γ_t from Equation (1), versus Dirichlet concentration α . Dots indicate means across five independent runs; shaded areas show the corresponding standard deviations.

Table 2: Final-round classifier-parameter gradient dissimilarity Γ_t as defined in Equation (1).

Dataset	FedAvg	FedProx	SCAFFOLD	FusedSpaceFed
Path	10.68	13.89	19.91	8.12
Derma	10.27	11.98	24.10	8.43
Retina	10.05	10.19	42.53	9.94
Blood	8.65	10.14	13.60	7.32

est measured final-round classifier gradient dissimilarity among the evaluated methods.

Contextual comparison with FedRep. Separately from the writer-level benchmark above, we evaluate FusedSpaceFed on a reconstructed ten-letter setting inspired by Collins et al. [5]: 150 synthetic clients, three classes per client, and 10% participation. With validation-selected learning rates, the sample-weighted test accuracy is $79.05 \pm 2.68\%$ over five seeds, averag-

Table 3: Accuracy in pathological partitioning.

Dataset	FedAvg	FedProx	SCAFFOLD	FusedSpaceFed
Path	20.49	19.66	17.16	39.62
Derma	25.75	24.43	22.46	48.15
Retina	34.37	33.02	26.61	49.26
Blood	18.22	19.48	18.10	30.60

ing rounds 191–200 within each run (sample standard deviation across runs). FedRep reports 78.56% for its FEMNIST setting. Appendix A details the experiment and compares with published results for FedRep and other methods.

5.4 Feature Heterogeneity on Digits

We further evaluate FusedSpaceFed on the five-domain Digits benchmark considered by Li et al. [24], comprising MNIST, SVHN, USPS, SynthDigits, and MNIST-M. Each domain defines one client, providing a com-

Table 4: **FEMNIST**. Writer-conditioned test performance and final-round classifier-gradient dissimilarity.

Method	Accuracy	Γ_t	F1-Score	Balanced Accuracy
FedAvg	85.49 $\pm$ 0.24	0.83 $\pm$ 0.01	66.87 $\pm$ 0.19	67.09 $\pm$ 0.20
FedProx	84.73 $\pm$ 0.18	0.81 $\pm$ 0.02	64.16 $\pm$ 0.15	65.81 $\pm$ 0.17
SCAFFOLD	84.51 $\pm$ 0.32	0.78 $\pm$ 0.02	65.68 $\pm$ 0.26	66.46 $\pm$ 0.30
FusedSpaceFed	86.82 $\pm$ 0.26	0.71 $\pm$ 0.02	68.77 $\pm$ 0.21	68.38 $\pm$ 0.19

mon ten-class recognition task with heterogeneous image characteristics across clients. Each client has 743 training examples, and all five clients participate in each of 300 communication rounds. With hyperparameters selected on a held-out validation split, FusedSpaceFed obtains a final-round test accuracy of $85.86 \pm 0.85\%$, averaged uniformly across domains and reported as mean $\pm$ sample standard deviation over five independent runs. Appendix C gives per-domain results, published FedAvg, FedProx and FedBN reference values, and the protocol.

Component analysis. How do the individual components contribute to the predictive performance of the complete fusion pipeline? We investigate reconstruction warm-up, encoder sharing, and the original-input pathway on Digits using five paired seeds and the same validation-selected hyperparameters. Retaining the original input improves mean uniform-domain accuracy by 1.00 percentage point relative to using the decoder output alone (85.86% versus 84.86%). Removing warm-up yields 85.71% , while sharing the encoder yields 86.18% . These comparisons support combining the original image with the learned decoder output in this feature-heterogeneous setting, with smaller mean changes under the other two interventions. See Appendix D.2 for full results and protocol.

5.5 Capacity and Computational Budget

Using the same Digits setting and the five FusedSpaceFed runs reported above, we additionally compare with a capacity- and compute-matched FedAvg variant. The FedAvg classifier is widened to closely match the number of active trainable parameters per client, and its local optimization budget is adjusted to match the final training FLOPs of FusedSpaceFed, including the reconstruction warm-up. Both methods use the same data partition, pre-processing and evaluation protocol, with hyperparameters selected separately on a common validation split. Across five paired seeds, the final-round test accuracy, averaged uniformly across the five domains, is $85.86 \pm 0.85\%$ for FusedSpaceFed and $82.06 \pm 0.22\%$ for FedAvg (mean $\pm$ sample standard deviation across runs). See Appendix B for details.

6 CONCLUSION

FusedSpaceFed combines private multiscale encoders with a shared decoder and classifier through additive input fusion. Local training alternates encoder-only reconstruction warm-up with end-to-end classification updates. Our exact fixed-round decomposition identifies when fusion-induced variance and cross-covariance reduce absolute classifier-gradient dispersion. Experiments show higher average predictive performance than the reference methods in most reported non-IID configurations and lower final-round classifier-gradient dispersion on the evaluated subset of datasets.

6.1 Current Limitations

FusedSpaceFed requires persistent private encoders; unseen-client adaptation remains untested.

It adds computation, memory and decoder communication, with validation-selected d_z .

Evaluation is limited to image tasks with convolutional autoencoders and CNN or MLP classifiers; the analysis concerns individual rounds.

6.2 Future Research Directions

Encoder distillation, progressive encoder alignment or encoder-free inference could enable a single deployable global model. Adaptive architectures could reduce costs in larger federations and support sustainable AI [2, 34].

Further directions include heterogeneous or multi-modal data, finite-sample analyses of joint model evolution, hierarchical or clustered federations, and evaluation under realistic communication and hardware constraints. Representation-level mechanisms could also be combined with adaptive optimization, including update rules selected using local gradients and curvature [7]. A preliminary PathMNIST experiment shows a substantial gain from federated refinement of the final classifier on frozen representations (Appendix E). Future work will assess robustness across seeds and datasets and the trade-off between accuracy and additional communication.

AI USE STATEMENT

In this work, we used generative AI tools to assist with manuscript editing, proofreading and revision; code cleanup, organization and debugging in preparation for the public repository; and bibliographic searches. We have reviewed all AI-assisted work. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes]

- (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes]
- (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes]
- (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

2. For any theoretical claim, check if you include:

- (a) Statements of the full set of assumptions of all theoretical results. [Yes]
- (b) Complete proofs of all theoretical results. [Yes]
- (c) Clear explanations of any assumptions. [Yes]

3. For all figures and tables that present empirical results, check if you include:

- (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes]
- (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
- (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes]
- (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes]

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:

- (a) Citations of the creator If your work uses existing assets. [Yes]
- (b) The license information of the assets, if applicable. [Not Applicable]
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- (d) Information about consent from data providers/curators. [Not Applicable]
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5. If you used crowdsourcing or conducted research with human subjects, check if you include:

- (a) The full text of instructions given to participants and screenshots. [Not Applicable]
- (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]

A RECONSTRUCTED FEMNIST COMPARISON WITH PUBLISHED BASELINES

A.1 Scope and data reconstruction

This experiment places FusedSpaceFed alongside the published FEMNIST results of Collins et al. [5]. We construct 150 synthetic clients from the ten lowercase letters a–j, with three classes per client.

We use the NIST Special Database 19 `by_class.zip` archive, specifically the `train_class` PNG directories. Images are converted to grayscale 28×28 inputs in $[0, 1]$, preserving orientation, with no augmentation. Following the historical generator’s recipe, each class pool contains at most 4,000 images. Client i , indexed from zero, receives classes $(i + j) \bmod 10$, $j = 0, 1, 2$, with requested per-class count $\lfloor (\exp Z_i + 100)/3 \rfloor$, where $Z_i \sim \mathcal{N}(4, 1)$; the data seed is 20261003.

The requested totals exceed the available pools for f (2,555 requested, 2,493 available), i (3,078; 2,788), and j (3,196; 1,920). For these classes only, we reduce each client’s quota proportionally and allocate integer remainders in decreasing fractional-remainder order, breaking ties by client ID. This changes the quotas of 105 clients. All assigned classes are retained, and each source image is assigned at most once globally. Each client is shuffled and split into $\lfloor 0.9N_i \rfloor$ training examples and the remaining test examples, using split seed 20261004. The resulting fixed partition contains 25,339 images: 22,736 training and 2,603 test, with disjoint source identifiers across clients and splits. All five runs use this same partition; variability across partitions is not measured.

A.2 Model and training choices

The classifier is an MLP with widths 784, 512, 256, 64, 10, biases, and ReLU after each hidden layer. The classifier dimensions follow the FedRep source code. We apply cross-entropy to logits.

FusedSpaceFed retains its private persistent encoder and shared decoder, using the compact U-Net with base width 16 and $d_z = 64$. The decoder has a linear output, and fusion is $x + D(E_i(x))$, without clamping or extra normalization. Each of 200 rounds samples 15 clients uniformly without replacement. With batch size 10, a selected client performs one epoch of encoder-only MSE warm-up, followed by five epochs of cross-entropy training of encoder, decoder, and classifier. The latter phase has no reconstruction loss. Decoder and classifier updates are averaged uniformly across selected clients.

The selected classifier optimizer is SGD with learning rate 0.02, momentum 0.5, weight decay 10^{-4} on weights and zero on biases. Encoder and decoder use Adam with learning rate 3×10^{-4} , $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, and no weight decay. Active gradients are clipped to L2 norm 1.0 separately for each optimizer. Optimizers restart at each client participation; Adam state persists between the two local phases. Learning rates are constant. Computation uses Float32 with AMP and TF32 disabled.

A.3 Validation-based calibration

Calibration uses only the original training portion. Within each client–class group, $\lfloor n/5 \rfloor$ examples are held out using a deterministic hash ordering with seed 20261004, leaving 18,374 fit and 4,362 validation examples. Six configurations are specified in advance, as triples (classifier learning rate, autoencoder learning rate, clipping norm): (0.01, 0.001, 1), (0.01, 0.0003, 1), (0.01, 0.0001, 1), (0.02, 0.0003, 1), (0.005, 0.0003, 1), and (0.01, 0.0003, 2).

All six are screened for 80 rounds with seed 141, using sample-weighted validation accuracy averaged over rounds 71–80. The reference and the two highest-scoring alternatives are extended to 200 rounds and also run with seed 142. Selection uses the mean of the two seeds’ validation accuracies over rounds 191–200. The chosen triple (0.02, 0.0003, 1) obtains 88.77% on this selection criterion. It is frozen before the five final runs, which start afresh with seeds 41–45 and use all 22,736 original training examples.

A.4 Evaluation and results

After each of rounds 191–200, all 150 clients are evaluated on their original local test examples, using the current shared decoder and classifier and their persistent private encoder, without test-time adaptation. For seed s and round t , let $c_{s,t,i}$ be the number of correct predictions on client i and n_i its test-set size. Our primary and secondary accuracies are, respectively,

$$A_{s,t} = 100 \frac{\sum_i c_{s,t,i}}{\sum_i n_i}, \quad U_{s,t} = \frac{100}{150} \sum_i \frac{c_{s,t,i}}{n_i}.$$

We average ten rounds within each seed, then report the mean and sample standard deviation ($n-1$ denominator) across five seed means. All runs complete 200 rounds; no best seed, round, or checkpoint is selected.

The primary result is $79.05 \pm 2.68\%$; the uniform-client result is $80.90 \pm 2.81\%$ (Table 5). Table 6 presents the comparison with the published FEMNIST reference values.

Table 5: Accuracy (%) by seed on our fixed reconstructed partition. Each entry averages rounds 191–200. SD is across five seed means.

Seed	Sample-weighted	Uniform-client
41	82.34	84.39
42	75.26	77.77
43	77.71	78.31
44	80.21	82.47
45	79.72	81.57
Mean	79.05	80.90
Sample SD	2.68	2.81

Table 6: Published FEMNIST reference values from Table 1 of Collins et al. [5], column (150, 3), and our separately measured result. Bold indicates the highest reported mean accuracy.

Method	Accuracy (%)
<i>Published; uncertainty not reported</i>	
Local Only	60.86
FedAvg	51.64
FedAvg+FT	72.41
FedProx	18.89
FedProx+FT	53.54
SCAFFOLD	17.65
SCAFFOLD+FT	52.11
Fed-MTL	54.11
PerFedAvg	71.51
LG-Fed	62.08
L2GD	66.18
APFL	70.74
Ditto	68.28
FedPer	76.91
FedRep	78.56
<i>Ours; reconstructed partition, mean $\pm$ SD</i>	
FusedSpaceFed, calibrated	79.05 $\pm$ 2.68

B CAPACITY- AND COMPUTE-MATCHED COMPARISON WITH FEDAVG

B.1 Digits setting and model

We use the five-domain Digits benchmark associated with Li et al. [24]: MNIST, SVHN, USPS, SynthDigits and MNIST-M. Each domain is assigned to one client with 743 training images, giving 3,715 training images in total. The fixed training partition and the full domain test sets are used for both methods. Images are represented as three-channel 28×28 inputs, with pixel values normalized to $[-1, 1]$. All five clients participate in each of 300 communication rounds, and shared states are aggregated uniformly.

The classifier has three convolutional layers with channel widths 3, 64, 64, 128, followed by fully connected layers with widths 6272, 2048, 512, 10. Its batch-normalization parameters and buffers are shared. FusedSpaceFed uses a compact U-Net with base width 16 and bottleneck width $d_z = 64$, a persistent private encoder for each client, and a shared decoder. Classification uses the fused input $x + D(E_i(x))$.

For FedAvg, the first fully connected layer is widened from 2048 to 2065 units; the following layer and the corresponding batch-normalization layer are adjusted accordingly. The active parameter count per client is

14,336,285 for FusedSpaceFed, comprising 14,219,210 classifier parameters, 72,080 encoder parameters and 44,995 decoder parameters. The enlarged FedAvg classifier has 14,334,589 parameters, a relative difference of 0.01183%. FusedSpaceFed stores its five client encoders separately, together with the shared decoder and classifier.

B.2 Local optimization and FLOP budget

Each FusedSpaceFed client performs one encoder-only reconstruction epoch with the decoder frozen, followed by one classification epoch updating the encoder, decoder and classifier. Reconstruction uses mean squared error; classification uses cross-entropy. The maximum batch size is 32, and the final batch of each epoch is retained.

The FedAvg local workload is set using the combined reconstruction and classification FLOP budget of FusedSpaceFed for each client and round. It comprises one pass through the local training set followed by additional shuffled minibatches. Batch sizes are selected using integer arithmetic, and the unspent remainder is carried to subsequent rounds.

Dense forward and backward operations are counted with Torch 2.5.1 `FlopCounterMode`, assigning two FLOPs to a fused multiply-add. The count includes gradient propagation through the frozen decoder dur-

ing reconstruction. Optimizer updates and gradient-norm clipping are accounted for using $5P$ scalar operations per SGD step and $16P$ per Adam step, where P is the number of parameters updated by that optimizer. Batch normalization, activations, pooling, loss evaluation, finite-value checks, copies, aggregation and kernel overhead are outside this FLOP accounting convention.

Table 7 gives the cumulative counts per final run. The relative difference in counted training FLOPs is 0.000198%. Across all clients and rounds, FusedSpaceFed performs 36,000 reconstruction steps and 36,000 classification steps; FedAvg performs 63,000 classification steps. The corresponding numbers of supervised sample presentations are 1,114,500 and 1,956,535.

Table 7: Active parameters per client and cumulative training FLOPs per run. The FusedSpaceFed FLOP count includes reconstruction warm-up.

Method	Parameters	FLOPs (10^{12})
FusedSpaceFed	14,336,285	551.898005
FedAvg, enlarged	14,334,589	551.896914

B.3 Validation-based calibration

For each domain, 149 of the 743 training images are assigned to validation and 594 to fitting, using deterministic class-stratified selection with data seed 20261005. The same split is used by both methods.

For FedAvg, nine configurations are evaluated for 60 rounds with seed 142, combining classifier learning rates $\{0.005, 0.01, 0.02\}$ and gradient-clipping norms $\{0.5, 1, 2\}$. Selection uses the uniform mean of domain validation accuracies at round 60 and yields classifier learning rate 0.02 and clipping norm 2.

FusedSpaceFed uses the validation-selected configuration described in Appendix C. Candidate configurations are screened for 120 rounds and shortlisted configurations are confirmed over 300 rounds using two validation seeds. The selected configuration uses classifier learning rate 0.1, autoencoder learning rate 3×10^{-4} , and clipping norm 10.

The classifier optimizer is SGD with zero momentum and weight decay. Encoder and decoder use Adam with $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and zero weight decay. Optimizers are initialized at each client participation, with Adam state retained between the two local phases. Learning rates are constant. Computation uses Float32 with AMP and TF32 disabled.

The selected configurations are fixed before the final runs. Both methods are trained from scratch on all

743 training images per client for 300 rounds, using paired seeds 42–46.

B.4 Evaluation and results

At round 300, each client is evaluated on its domain’s full test set. FusedSpaceFed predictions combine that client’s encoder with the final shared decoder and classifier; FedAvg uses its final shared classifier. Evaluation uses inference mode and the parameters obtained from training.

Let $c_{s,d}$ and n_d denote the correct-prediction count and test-set size for seed s and domain d , respectively. The uniform-domain primary accuracy and the sample-weighted accuracy are, respectively,

$$U_s = \frac{100}{5} \sum_{d=1}^5 \frac{c_{s,d}}{n_d}, \quad A_s = 100 \frac{\sum_{d=1}^5 c_{s,d}}{\sum_{d=1}^5 n_d}.$$

We report means and sample standard deviations across the five paired seeds, using the $n - 1$ denominator. The uniform-domain accuracy is $85.86 \pm 0.85\%$ for FusedSpaceFed and $82.06 \pm 0.22\%$ for FedAvg. Table 8 reports the primary result and paired differences. The sample-weighted result is $84.51 \pm 0.75\%$ for FusedSpaceFed and $79.68 \pm 0.39\%$ for FedAvg. Table 9 gives the domain-level results.

Table 8: Final-round uniform-domain accuracy (%) and paired difference (FusedSpaceFed minus FedAvg, in percentage points). Bold indicates the higher accuracy in each row.

Seed	FusedSpaceFed	FedAvg	Difference
42	84.67	82.27	+2.40
43	86.30	82.15	+4.15
44	85.32	82.25	+3.08
45	86.76	81.78	+4.98
46	86.26	81.89	+4.37
Mean	85.86	82.06	+3.80
Sample SD	0.85	0.22	1.04

Table 9: Final-round domain accuracy (%): mean and sample standard deviation across five seeds. Bold indicates the higher mean accuracy in each row.

Domain	FusedSpaceFed	FedAvg
MNIST	96.95 $\pm$ 0.24	95.93 $\pm$ 0.24
SVHN	67.83 $\pm$ 2.88	61.57 $\pm$ 1.43
USPS	96.67 $\pm$ 0.28	95.94 $\pm$ 0.25
SynthDigits	86.32 $\pm$ 0.50	81.31 $\pm$ 0.54
MNIST-M	81.55 $\pm$ 0.77	75.58 $\pm$ 0.64

Table 10: Test accuracy (%) on the five-domain Digits benchmark. Published reference values for FedAvg, FedProx, and FedBN are taken from Table 11 of the supplementary material of Li et al. [24]. FusedSpaceFed results are mean $\pm$ sample standard deviation over five independent runs, evaluated at communication round 300. Bold indicates the highest reported mean accuracy in each row.

Domain	FedAvg published	FedProx published	FedBN published	FusedSpaceFed ours
MNIST	97.38 $\pm$ 0.05	97.30 $\pm$ 0.17	97.55 $\pm$ 0.11	96.95 $\pm$ 0.24
SVHN	70.59 $\pm$ 0.51	71.55 $\pm$ 0.75	76.93 $\pm$ 0.25	67.83 $\pm$ 2.88
USPS	96.91 $\pm$ 0.11	96.98 $\pm$ 0.19	97.69 $\pm$ 0.10	96.67 $\pm$ 0.28
SynthDigits	86.66 $\pm$ 0.21	86.60 $\pm$ 0.18	87.46 $\pm$ 0.20	86.32 $\pm$ 0.50
MNIST-M	82.44 $\pm$ 0.41	82.67 $\pm$ 0.75	83.57 $\pm$ 0.38	81.55 $\pm$ 0.77

C FEATURE HETEROGENEITY ON DIGITS

We use the data partition, preprocessing, and FusedSpaceFed architecture described in Appendix B. Each domain contributes 743 training examples, with the class frequencies present in the distributed partition. Images are represented as three-channel 28×28 inputs normalized to $[-1, 1]$. All five clients participate in every round, with uniform aggregation of the shared decoder and classifier. The classifier’s batch-normalization parameters and running statistics are included in its shared state.

Hyperparameter selection uses a fixed split of 594 training and 149 validation examples per client. Candidate configurations are screened for 120 rounds, followed by 300-round confirmation of shortlisted configurations using two validation seeds. Selection is based on the mean final-round validation accuracy across domains and confirmation seeds. The selected configuration uses classifier SGD with learning rate 0.1, encoder–decoder Adam with learning rate 0.0003, and gradient-norm clipping at 10. Each communication round includes one encoder-only reconstruction warm-up epoch followed by one end-to-end classification epoch, with batch size 32.

The selected configuration is then trained from scratch on all 743 training examples per client using seeds 42–46. At round 300, each client-conditioned predictor is evaluated on its domain’s test set. We report the mean and sample standard deviation across the five runs for each domain in Table 10. The uniform mean across domains is computed within each run and then summarized across runs.

Unlike our severe label-skew experiments, the Digits benchmark primarily evaluates feature heterogeneity, which FedBN is specifically designed to address. This difference in regime may partly explain the gap to the published FedBN results, although validating this hypothesis would require further controlled experiments.

D COMPONENT AND GRADIENT DIAGNOSTICS ON DIGITS

We use the five calibrated FusedSpaceFed runs (seeds 42–46) reported in Sections 5.4 and 5.5. The data partition, architecture, and validation-selected hyperparameters are those in Appendices B and C. The full-model reference is therefore the same $85.86 \pm 0.85\%$ uniform-domain accuracy reported in the main text.

D.1 Decoder Behavior and Reconstruction Warm-Up

What role does reconstruction warm-up play before task-driven optimization? We replay one local participation from copies of the initial and final (round 300) checkpoints of the five full-model runs, using a fixed training probe of 160 examples per domain, arranged in five batches of 32. Each replay performs one encoder-only warm-up epoch followed by one end-to-end classification epoch. Optimizers are reset at the start of participation, as in the Digits training protocol, with continuous Adam state between the two phases. Metrics are evaluated with frozen BatchNorm statistics. The intermediate states are local diagnostic replays from the saved checkpoints.

Warm-up reduces reconstruction MSE averaged across seeds in every domain at both checkpoint stages (Table 11). At the final checkpoints, MSE averaged across domains and seeds decreases from 0.4512 to 0.4477 after warm-up, then increases to 0.5028 after classification. These observations illustrate the distinct roles of the two phases: warm-up improves reconstruction, whereas subsequent classification shapes the decoder output through the prediction objective alone.

The decoder ends with a linear convolution, allowing output values beyond the normalized input range $[-1, 1]$. Figure 2 illustrates the image-space signals before and after each phase. All measured outputs, losses, and model states remain finite in these replays. During warm-up, only the encoder changes; classification updates the encoder, decoder, and classifier.

D.2 Component Ablations

How do the design choices affect the accuracy of the complete fusion pipeline? We reuse the five full-model runs and train each of three variants from scratch with the same five seeds, giving 15 additional runs of 300 rounds. The variants remove reconstruction warm-up, share the encoder, or remove the original-input pathway. In the shared-encoder variant, all clients start with the same encoder and uniformly average its state together with the decoder and classifier after each round. The decoder-only variant classifies $D(E_i(x))$

while retaining private encoders and warm-up.

All variants inherit the hyperparameters selected for the full pipeline, without additional tuning: classifier SGD with learning rate 0.1, encoder-decoder Adam with learning rate 0.0003, clipping at 10, batch size 32, and $d_z = 64$. Training uses all 743 examples per domain and one classification epoch per round; the full, shared-encoder, and decoder-only variants additionally perform one warm-up epoch. Each final model is evaluated once on the five domain test sets, containing 147,509 examples in total. Uniform-domain accuracy is computed within each run.

Results are reported as mean $\pm$ sample standard deviation across five paired seeds; differences are computed within each seed. Removing warm-up also reduces the optimization budget and changes the Adam state and minibatch order at the start of classification. This intervention therefore measures the overall effect of removing the warm-up phase.

The complete pipeline achieves 85.86% mean uniform-domain accuracy, compared with 84.86% when the original-input pathway is removed. This 1.00-point average advantage supports combining the learned decoder output with the original image. The no-warm-up and shared-encoder variants obtain 85.71% and 86.18%, respectively, showing smaller mean changes under these interventions in the feature-heterogeneous Digits setting.

D.3 Fixed-State Gradient Dispersion

Can the learned fusion pathway realize the variance-covariance reduction condition in Equation (3)? We compare original and fused inputs at the initial and final checkpoints using identical classifier, decoder, and private-encoder parameters. We use the same fixed training probes as in Appendix D.1, with no optimizer steps or adaptation. For each client, we average the gradients of five minibatch losses before computing cross-client dispersion.

The primary measurement freezes BatchNorm running statistics. A second measurement uses batch statistics and restores all buffers before and after each input pathway. These batch-statistic measurements condition the loss on the fixed minibatches. Gradients are computed in float32 and accumulated in float64. Table 13 reports the terms of the decomposition, which satisfy the identity to numerical precision.

At the final checkpoints, the mean per-seed ratio $\Gamma^{\text{fused}}/\Gamma^{\text{orig}}$ is 0.792 ± 0.253 with frozen BatchNorm statistics and 0.148 ± 0.141 with batch statistics. The reduction condition holds in four of five seeds and all five seeds, respectively. These trained fusion pathways

Table 11: Reconstruction MSE on the fixed Digits training probes, averaged over five seeds. Each checkpoint is copied before a local replay. Before, Warm-up, and Classification denote successive replay states; the two column groups start from initialization and round 300.

Domain	Initial checkpoints			Final checkpoints		
	Before	Warm-up	Classification	Before	Warm-up	Classification
MNIST	0.9428	0.9213	0.9239	0.9023	0.8962	1.1163
SVHN	0.1965	0.1921	0.1924	0.1877	0.1853	0.1955
USPS	0.5946	0.5804	0.5813	0.5649	0.5611	0.5715
SynthDigits	0.3730	0.3656	0.3670	0.3517	0.3482	0.3668
MNIST-M	0.2845	0.2795	0.2772	0.2496	0.2475	0.2639

Table 12: Digits component ablations. Uniform-domain test accuracy at round 300 and paired differences are mean $\pm$ sample standard deviation over seeds 42–46. Positive differences favor the full pipeline. Bold marks the highest mean accuracy among the evaluated variants.

Variant	Uniform-domain accuracy (%)	Full – variant (pp)
Full FusedSpaceFed	85.86 $\pm$ 0.85	–
Without warm-up	85.71 $\pm$ 0.79	+0.15 $\pm$ 0.53
Shared encoder	86.18 $\pm$ 0.47	–0.32 $\pm$ 0.58
Decoder output only	84.86 $\pm$ 0.58	+1.00 $\pm$ 1.22

therefore provide empirical instances of the absolute-dispersion reduction identified by the analysis.

To distinguish absolute dispersion from gradient scale and direction, we also measure $\Gamma/(K^{-1} \sum_i \|g_i\|_2^2)$ and the mean pairwise cosine similarity of client gradients (Table 14). With frozen statistics, final-state normalized dispersion changes from 0.7942 to 0.7996 and mean pairwise cosine similarity from 0.0906 to 0.0440. With batch statistics, the corresponding values change from 0.7966 to 0.7890 and from 0.0218 to 0.0235. These results characterize a reduction in absolute dispersion, with substantial sensitivity to gradient scale and BatchNorm statistics. Both input pathways are evaluated at the same FusedSpaceFed state, whose classifier was trained on fused inputs. Thus, the original-input pathway is a fixed-state counterfactual; the cross-method measurements in Section 5 instead use each method’s own final state.

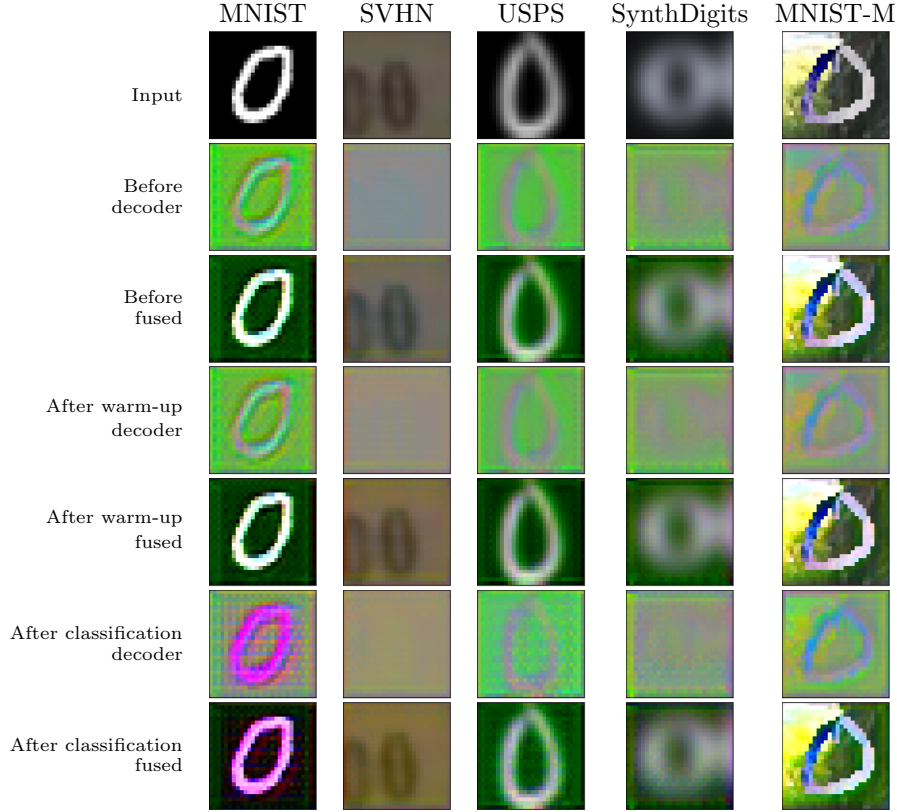


Figure 2: Digits decoder probes from the final checkpoint, seed 42. Each column uses the first preselected class-0 example in that domain; the same example is shown throughout its local replay. Rows show the input and the decoder and fused outputs before adaptation, after warm-up, and after classification. All panels retain the fixed display range $[-1, 1]$ of the archived grids; clipping is applied only for visualization, without panel-specific rescaling. Quantitative measurements use the raw, unclipped outputs.

Table 13: Fixed-state classifier-gradient decomposition on Digits. Terms Γ^{orig} , Γ^{fused} , B , and Φ are means across five seeds. The ratio is computed within each seed and reported as mean $\pm$ sample standard deviation. The last column counts seeds satisfying $B + \Phi < 0$. Frozen and Batch denote the two BatchNorm modes.

Stage	BN	Γ^{orig}	Γ^{fused}	B	Φ	$\Gamma^{\text{fused}}/\Gamma^{\text{orig}}$	Reduced
Initial	Frozen	2.001×10^{-2}	2.012×10^{-2}	1.054×10^{-3}	-9.383×10^{-4}	1.006 ± 0.013	2/5
Initial	Batch	78.6571	78.5144	10.5506	-10.6934	0.998 ± 0.004	3/5
Final	Frozen	13.8595	10.6770	1.2951	-4.4775	0.792 ± 0.253	4/5
Final	Batch	6.732×10^{-2}	1.580×10^{-3}	5.700×10^{-2}	-0.1227	0.148 ± 0.141	5/5

Table 14: Normalized and directional Digits gradient diagnostics, averaged across five seeds. Normalized dispersion is $\Gamma/(K^{-1} \sum_i \|g_i\|_2^2)$; cosine similarity is averaged over distinct pairs of client gradients. Orig. and Fused use identical model parameters and fixed training probes.

Stage	BN	Normalized dispersion		Pairwise cosine	
		Orig.	Fused	Orig.	Fused
Initial	Frozen	0.7989	0.7988	-0.0048	-0.0049
Initial	Batch	0.7364	0.7370	0.0775	0.0763
Final	Frozen	0.7942	0.7996	0.0906	0.0440
Final	Batch	0.7966	0.7890	0.0218	0.0235

E FEDERATED CLASSIFIER REFINEMENT ON PATHMNIST

Classifier calibration after federated training has previously been studied under data heterogeneity [26]. We explore this direction for FusedSpaceFed by refining its final linear classification layer while keeping the learned representation fixed.

We start from the original round-50 PathMNIST checkpoint for seed 42, trained with mixed precision using ten clients and two classes per client. Private encoders, the shared decoder, the classifier backbone, and all BatchNorm buffers remain unchanged. Only the 585 parameters of the final 64-to-9 linear layer are optimized, starting from their checkpoint values.

In a federated simulation with separate client processes, each client computes frozen features from its own training images using its private encoder. The server broadcasts candidate head parameters and receives only local losses and head gradients. Uniform averaging across clients implements the mean client cross-entropy objective without pooling features or labels. Optimization uses FP32 L-BFGS with learning rate 1, strong-Wolfe line search, history size 20, and a fixed budget of 100 iterations. The refined head is frozen before test evaluation.

Table 15: PathMNIST head refinement from the same original checkpoint (seed 42, round 50). Accuracy is averaged uniformly over ten client-conditioned predictors, each evaluated on the same 7,180 official test images. All entries concern a single seed.

Model	Accuracy (%)
Original FusedSpaceFed checkpoint	39.62
Centralized head refinement, pooled training features	75.33
Federated head refinement, client-local training features	75.46

Federated refinement improves accuracy by 35.84 percentage points (Table 15). It requires 107 additional full-participation gradient aggregations, corresponding to 5.01 MB of bidirectional optimization payload, excluding initial model distribution and transport overhead.

Loss and gradient checks at common parameter values satisfy the prespecified tolerances against the centralized reference. The two truncated optimization runs nevertheless produce different terminal models, with 2.36% disagreement in test predictions, and do not satisfy the stricter terminal-equivalence criteria.

This experiment shows that substantial predictive performance can be recovered through final-layer optimization while preserving client-local data processing. The single-seed result motivates further evaluation of this additional federated stage across seeds, datasets, and communication budgets.